[INCH-POUND]

MIL-R-914/4
12 February 1991

MILITARY SPECIFICATION

RESISTOR NETWORK, SURFACE MOUNT,
20-PIN, LEADLESS CHIP CARRIER,
ESTABLISHED RELIABILITY, STYLE RNS040

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. The specification covers the detail requirements for style RNS040 resistor networks. This style is available in all characteristics and all resistance tolerances.

1.2 Part or Identifying Number (PIN) (see 6.4). Resistor networks covered by this specification are identified by a PIN which consists of the basic number of this specification and a coded number. The military PIN is in the following form:



The coded number is derived in accordance with MIL-R-914.

1.2.1 Schematic. The schematic of the resistor network is identified by a single letter in accordance with figure 1. The resistor element R_{ref} is the reference resistor element used in determining the ratio accuracy (when applicable).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

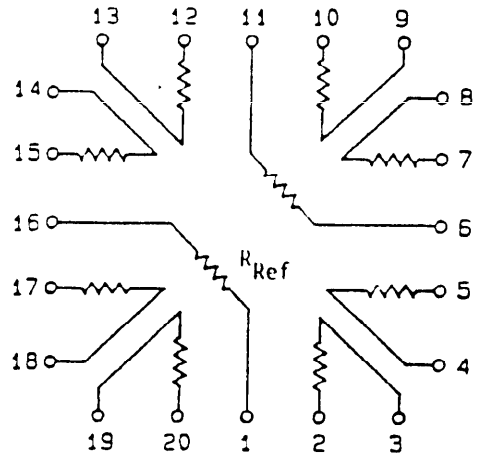
2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

Benefical comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Laboratory Command, ATTN: SLCET-R-S, Fort Monmouth, NJ 07703-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the of this document or by letter.

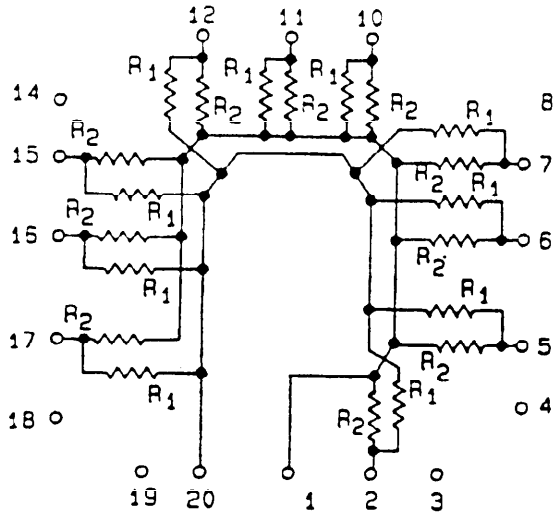
AMSC N/A FSC 5905

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

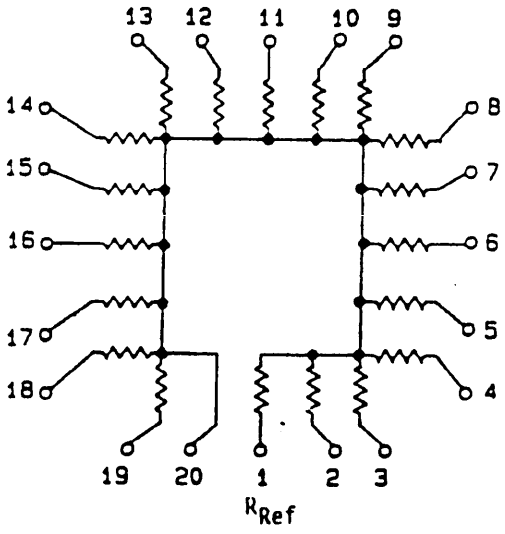
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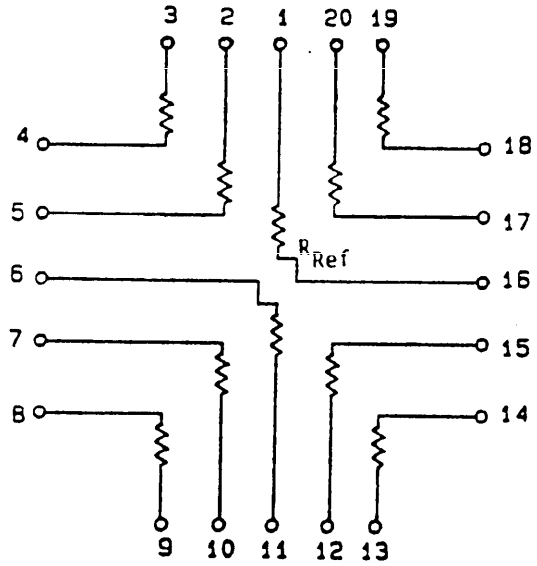
SCHEMATIC P



SCHEMATIC S



SCHEMATIC M



SCHEMATIC E

FIGURE 1. Schematics.

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SPECIFICATIONS

MILITARY

MIL-R-914 - Resistor Networks, Fixed, Film, Surface Mount, Established Reliability, General Specification for.
MIL-R-39032 - Resistors, Packaging of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Requirements. Requirements shall be in accordance with MIL-R-914 and as specified herein.

3.2 Design and construction. The resistor networks shall be of the design, construction, and physical dimensions specified on figure 2.

3.3 Power rating. The power ratings for schematics P, M, E, and S shall be as specified in table I.

TABLE I. Power ratings.

Schematics	Characteristics K and M		Characteristics C, R, H, V	
	Element (watts)	Network (watts)	Element (watts)	Network (watts)
P	0.10	1.00	0.050	0.50
M	0.05	0.95	0.025	0.50
E	0.10	1.00	0.050	0.50
S	0.05	1.00	0.025	0.50

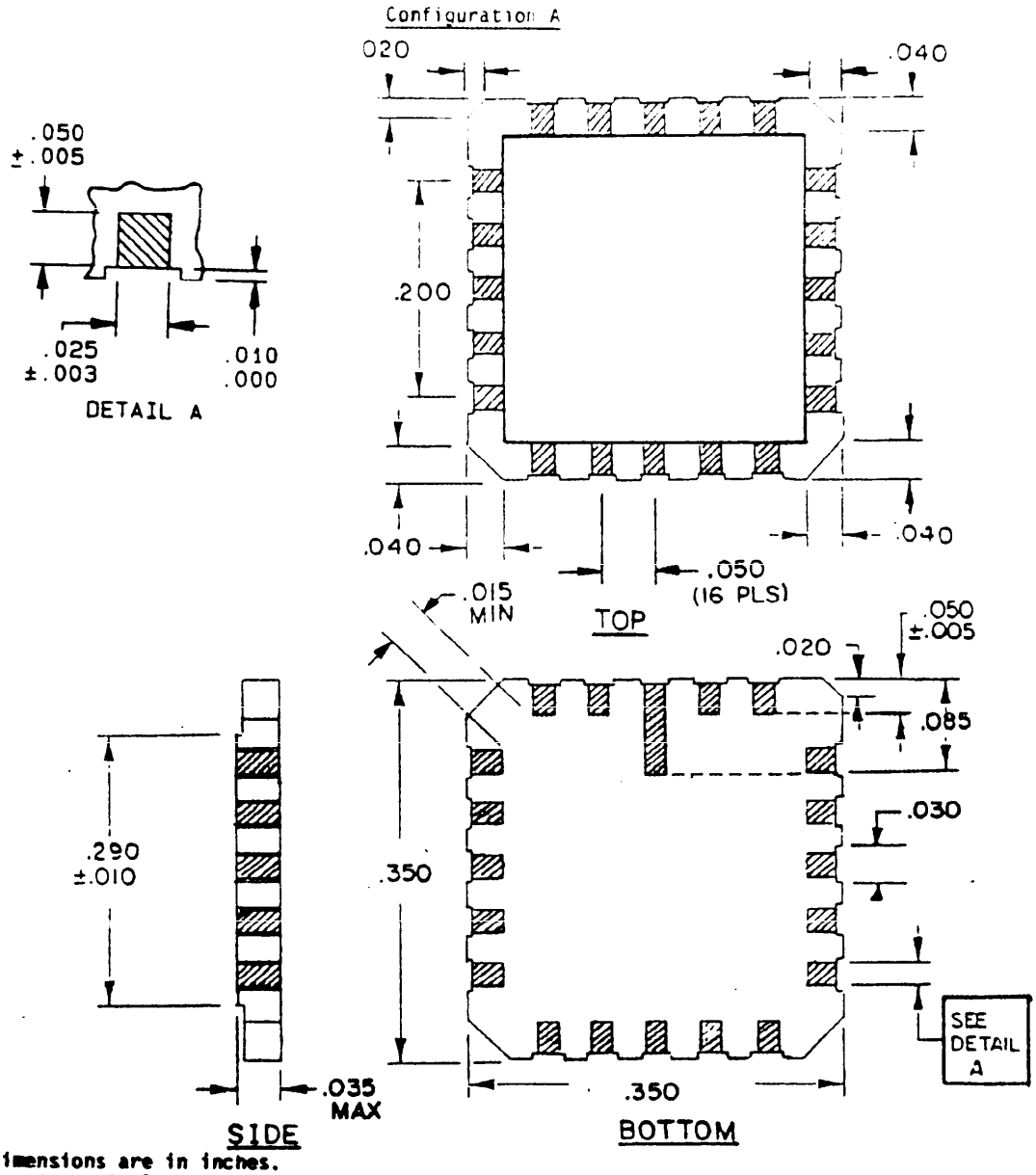
3.4 Power conditioning. The power applied for power conditioning for schematics P, M, E, and S shall be 1.5 times rated power.

3.5 Temperature range. The operating temperature range shall be from -55°C to +125°C.

3.6 Voltage rating. The maximum voltage rating shall be 50 V dc.

3.7 Resistance range. The resistance range shall be 10 ohms to 2.2 megohms.

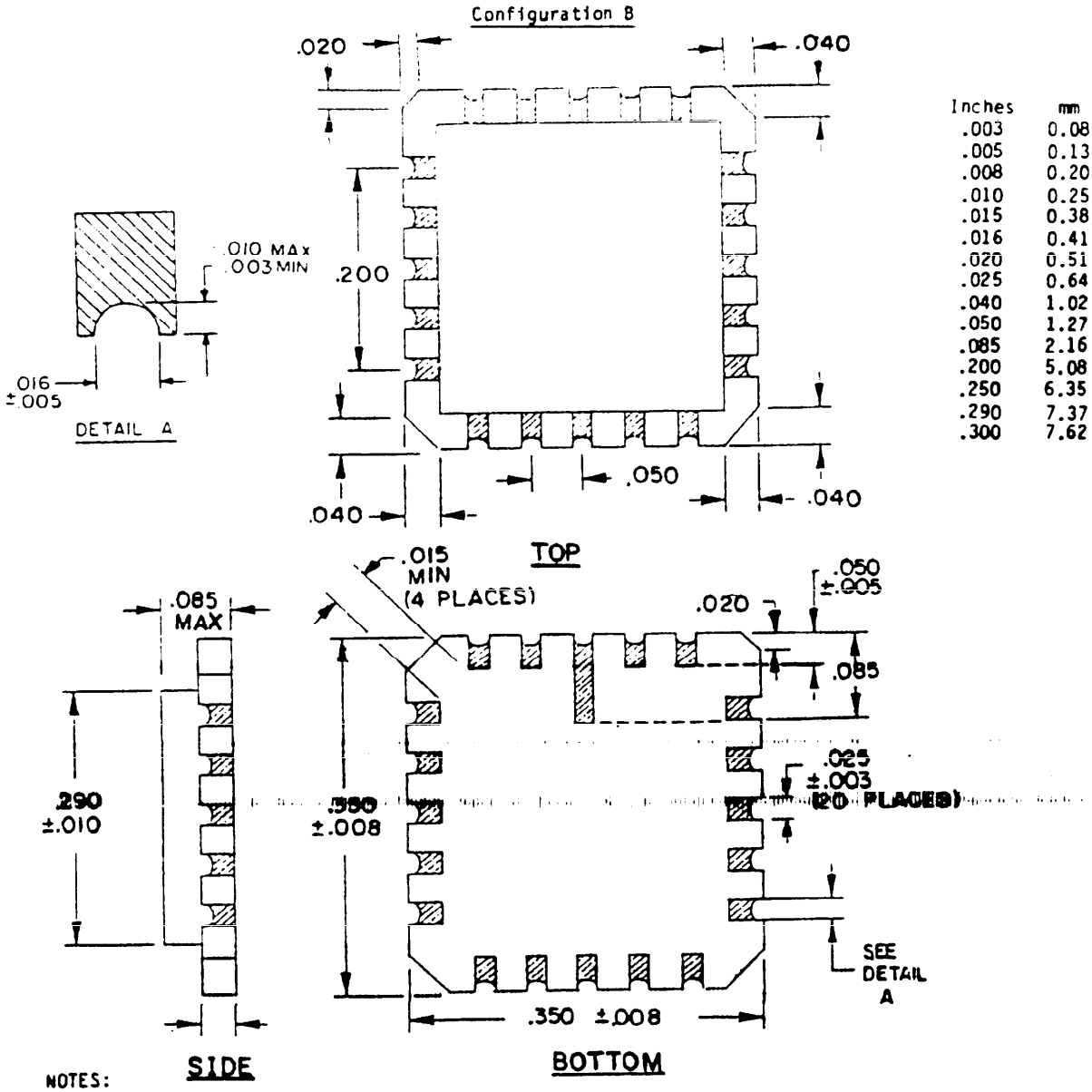
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NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.008 (0.20 mm).
4. Adjacent corner pads may be rounded or diagonally cut to meet the .015 (.038 mm) minimum requirement.
5. Covers termination materials A, D, and G.

FIGURE 2. Leadless chip carrier.

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Inches	mm
.003	0.08
.005	0.13
.008	0.20
.010	0.25
.015	0.38
.016	0.41
.020	0.51
.025	0.64
.040	1.02
.050	1.27
.085	2.16
.200	5.08
.250	6.35
.290	7.37
.300	7.62

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances are ±.008 (0.20 mm).
4. Adjacent corner pads may be rounded or diagonally cut to meet the .015 (.038 mm) minimum requirement.
5. Covers termination materials B, E, and H.

FIGURE 2. Leadless chip carrier - Continued.

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3.8 Standard resistance values. The standard resistance values and the resistance designators for the "S" schematic are as specified in table II.

TABLE II. Standard resistance values.

Resistance designator	R1 (ohms)	R2 (ohms)	Resistance designator	R1 (ohms)	R2 (ohms)
A001	82	130	A010	330	470
A002	120	200	A011	330	680
A003	130	210	A012	1.5K	3.3K
A004	160	260	A013	3K	6.2K
A005	180	240	A014	180	270
A006	180	390	A015	270	270
A007	220	270	A016	560	560
A008	220	330	A017	560	1.2K
A009	330	390	A018	620	2.7K

3.9 Resistance tolerance. Resistors are available in (B) ±0.1 percent, (D) ±0.5 percent, (F) ±1.0 percent, (G) ±2 percent, (J) ±5 percent resistance tolerances.

3.10 Marking. Due to the size limitations, this style resistor shall be marked with the following minimum information:

042K - Date code, characteristic.
10K0 - Coded resistance value.
FMJ - Tolerance, failure rate, JAN marking.

3.11 Terminations. Resistors are available in terminations A, D, and G for configuration A and terminations B, E, and H for configuration B.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-R-914.

4.2 "S" schematic tests. For the tests on "S" schematic, the dc resistance measurements, resistance temperature characteristic measurement, and short time overload application shall be made between each terminal and applicable common terminal without compensation for the shunt circuits to determine the acceptability to section 3 requirements.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements shall be in accordance with MIL-R-39032.

5.2 Marking. Package shall be marked with assigned PIN, manufacturer's name or logo, code number (CAGE) and date and lot codes.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not manatory.)

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6.1 Intended use. Resistor networks are used in surface mounting applications where space is a major concern.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Complete PIN.

6.3 Soldering. Extreme care should be taken when soldering these resistors. Prolonged exposure to high temperatures can damage these devices.

6.4 Part or Identifying Number (PIN) (see 1.2). PIN is a new term encompassing terms previously used in specifications such as part number, type designator, identification number, etc.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85
NASA - NA

Review activities:
Army - AR, MI
Navy - AS, OS
Air Force - 17, 85
DLA - ES

User activities:
Army - AT, AV, ME
Navy - CG, MC
Air Force - 19

Preparing activity:
Army - ER

Agent:
DLA - ES

(Project 5905-1143-4)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-R-914/4	2. DOCUMENT DATE (YYMMDD) 91/02/12

3. DOCUMENT TITLE
Resistor Network, Fixed, Film, 20-Pin, Established Reliability, Style RNS040

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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temperature of any other resistor. There shall be no circulation of air directly over the